



CONTROL SYSTEMS

Faculty team

RRC:

Dr. Anuradha M – Anchor

Dr. Aswini N, Dr. Rashmi N.U

Dr. Anagha K

ECC:

Prof. Muralidhar S

EEE:

Prof. K.R.Pushpa

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Signal Flow Graph



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

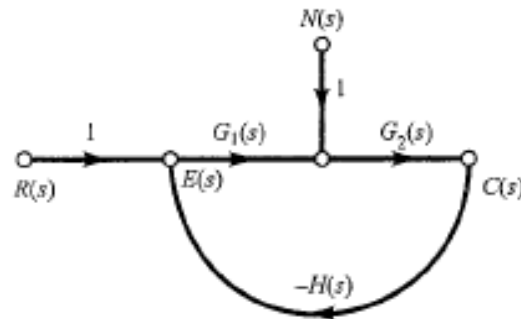
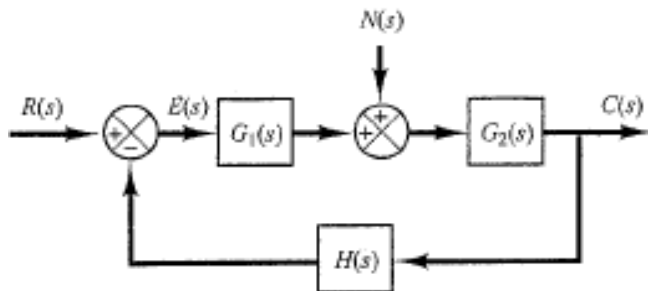
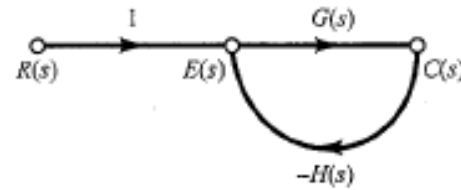
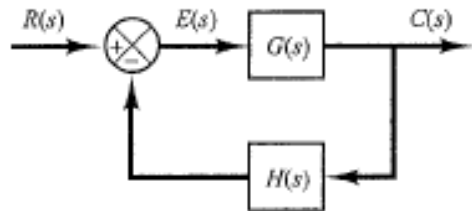
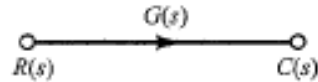
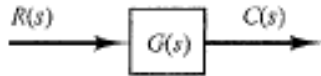


- An alternative method for determining the relationship between system variables has been developed by **S. J. Mason**.
- Adv: This method does not require any reduction techniques as such because of availability of Mason's gain formula
- A **signal-flow graph** is a diagram consisting of nodes that are connected by several directed branches and is a graphical representation of a set of linear relations.

$$Y(s) = G(s) R(s)$$
A hand-drawn signal flow graph in red ink. It consists of two nodes, represented by small circles. The input node on the left is labeled $R(s)$ and the output node on the right is labeled $Y(s)$. A directed branch with an arrow points from the input node to the output node, and this branch is labeled with the transfer function $G(s)$.

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Block Diagram to Signal Flow Graph

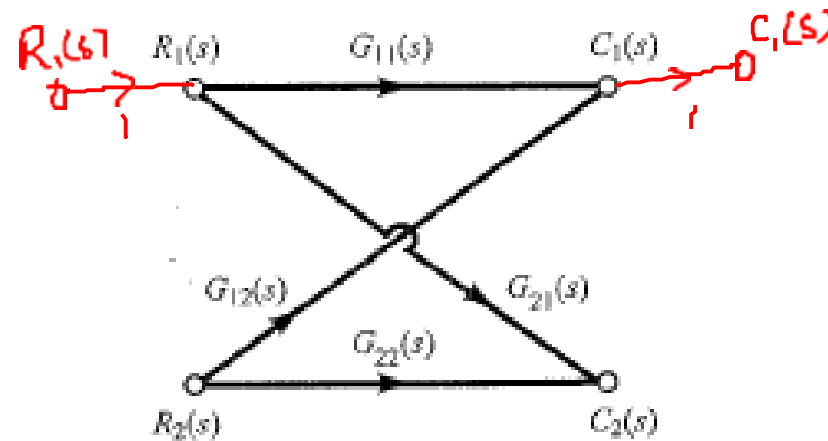
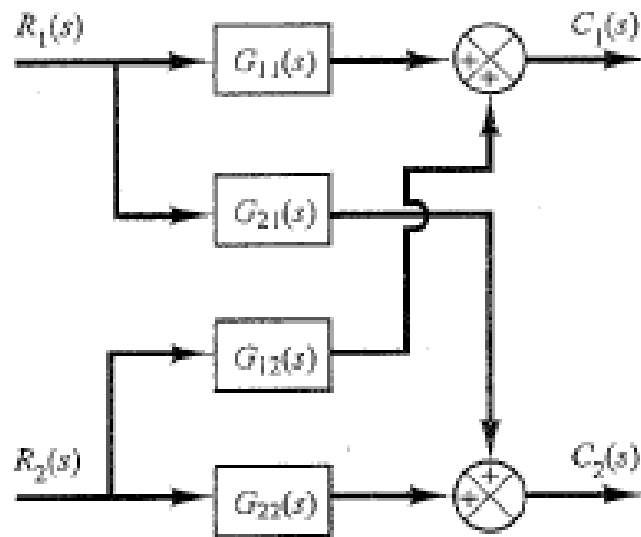
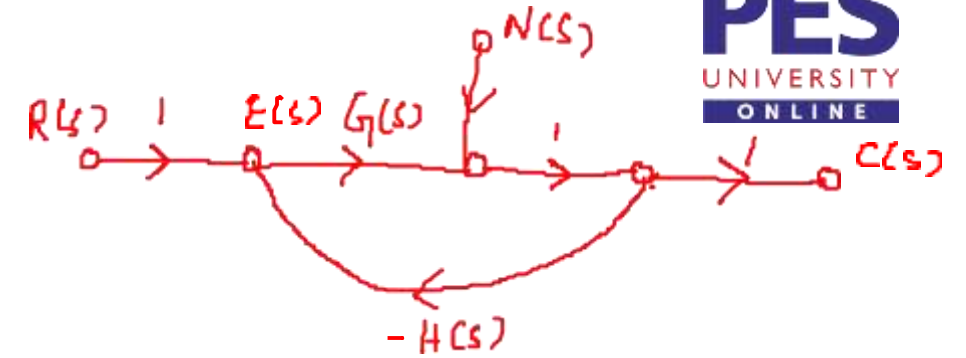


MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Block Diagram to Signal Flow Graph



PES
UNIVERSITY
ONLINE

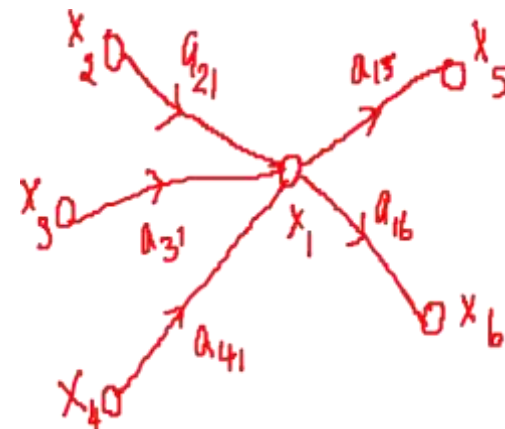
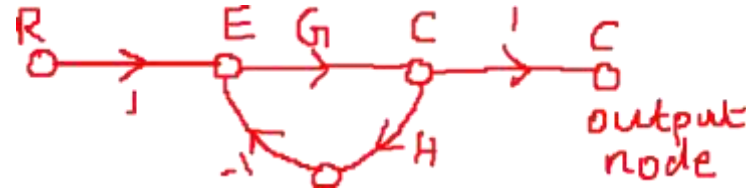


MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

Terminology:

- **Branch:** A signal travels along a branch from one node to another in the direction indicated by the branch arrow and in the process gets multiplied by the gain or transmittance of the branch.
- For ex, signal traversing from node E to node C is given by GE



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

Terminology:

- **Nodes:** The input and output points or junctions are called nodes.

- **Node as a summing point:** sum of all incoming signals $x_2, x_3, x_4, x_1, x_5, x_6$

$$x_1 = a_{21}x_2 + a_{31}x_3 + a_{41}x_4$$

- **Node as a transmitting point:** transmitted through all branches outgoing from the node

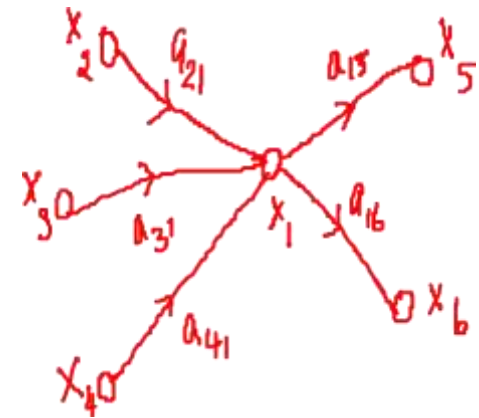
$$x_5 = a_{15}x_1$$

$$x_6 = a_{16}x_1$$

- **Input node or source node:** node with only outgoing branches
- **Output node or sink:** node with only incoming branches

$$x_2, x_3, x_4$$

$$x_5, x_6$$

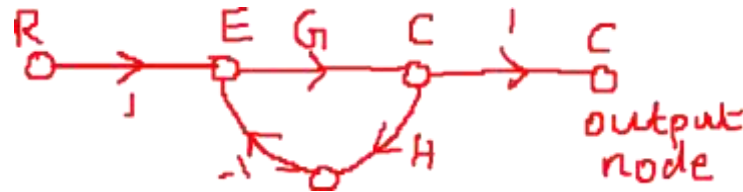


MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

Terminology:

- **A path** is a branch or a continuous sequence of branches that can be traversed from one signal (node) to another signal (node).
- **Forward path:** It is a path from the input node to the output node

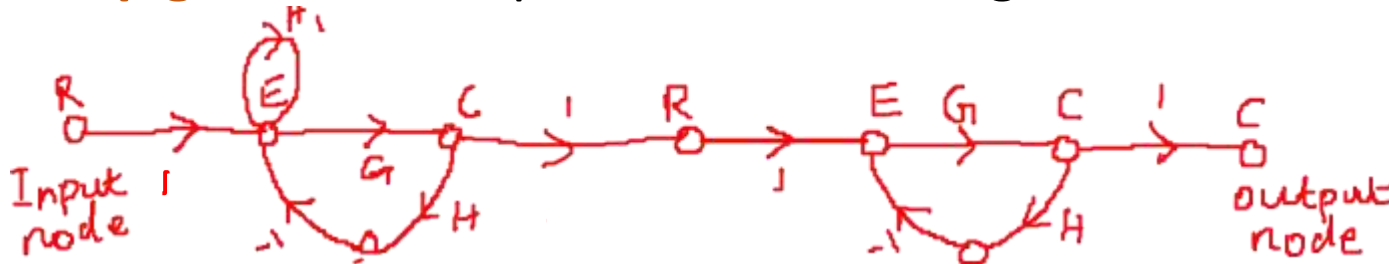


MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

Terminology:

- **A loop** is a closed path that originates and terminates on the same node, with no node being met twice along the path.
- **Non-touching loops:** Two loops are said to be non-touching if they do not have a common node.
- **Forward path gain:** product of the branch gains , G_1
- **Loop gain:** It is the product of branch gains encountered in traversing the loop $-G_1H$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

Construction of Signal Flow Graph:

Signal-Flow Graph Models

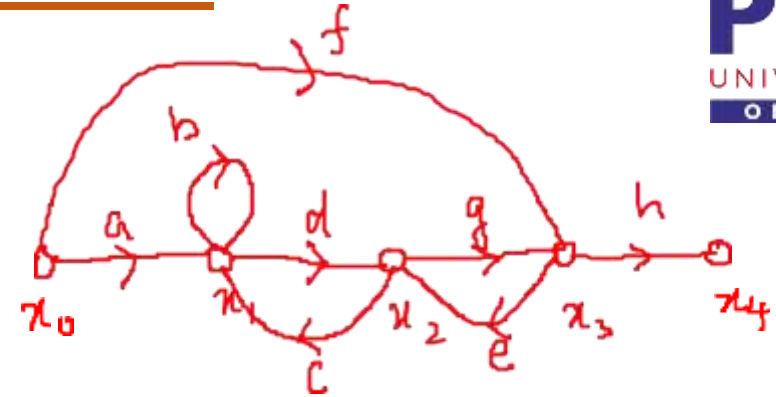
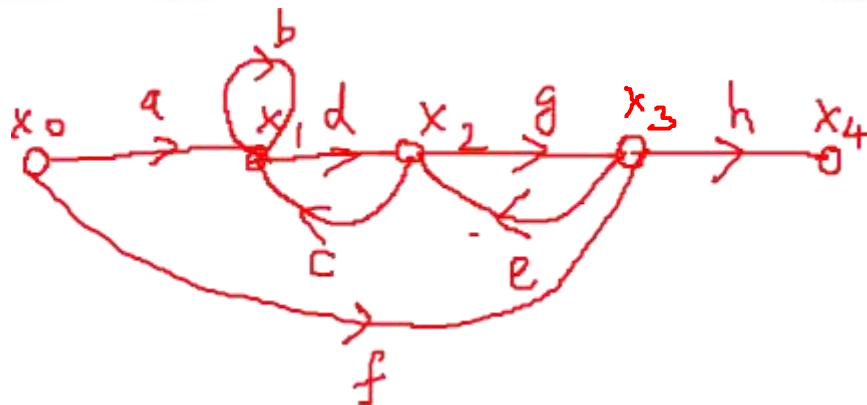
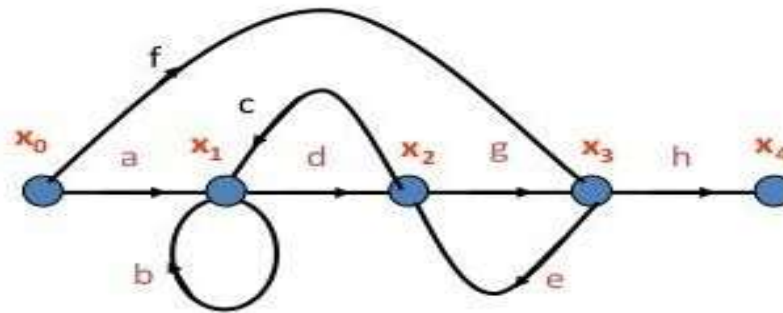
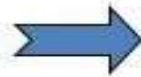
x_0 is input and x_4 is output

$$x_1 = ax_0 + bx_1 + cx_2$$

$$x_2 = dx_1 + ex_3$$

$$x_3 = fx_0 + gx_2$$

$$x_4 = hx_3$$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

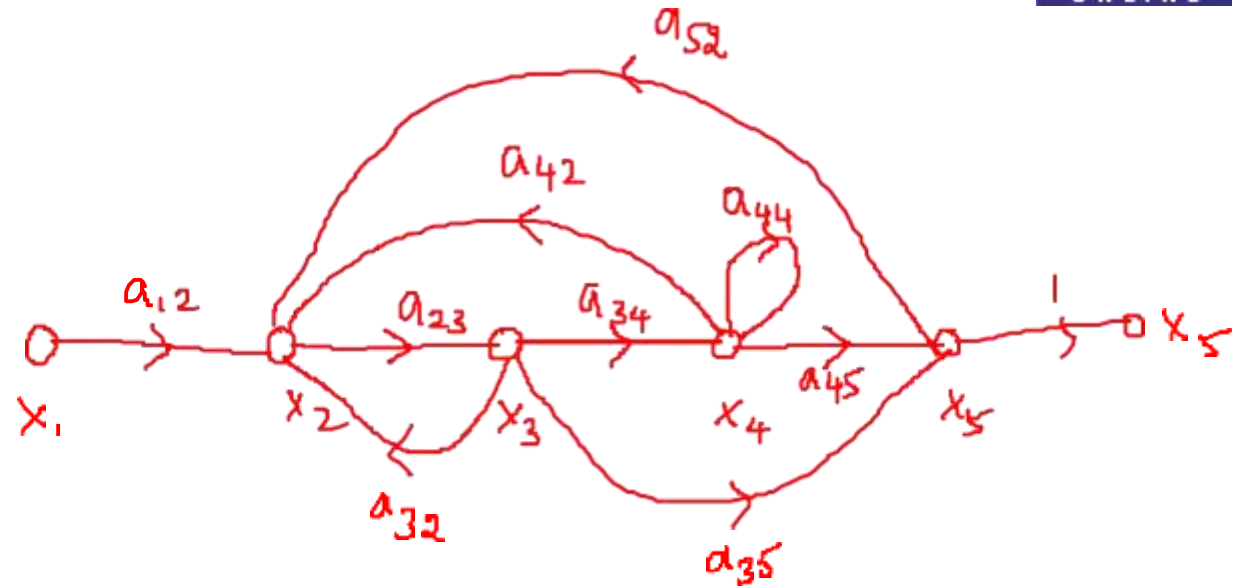
Construction of Signal Flow Graph:

$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$

$$x_3 = a_{23}x_2$$

$$x_4 = a_{34}x_3 + a_{44}x_4$$

$$x_5 = a_{35}x_3 + a_{45}x_4$$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph

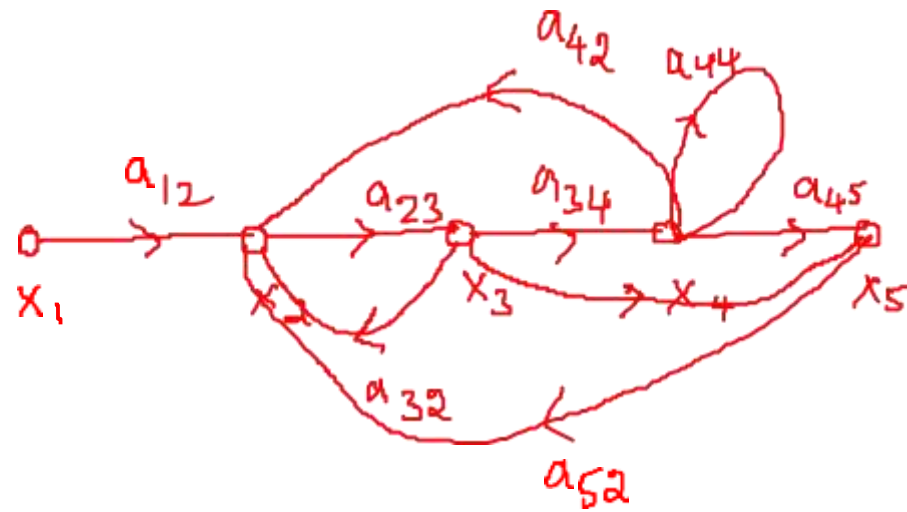
Construction of Signal Flow Graph:

$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$

$$x_3 = a_{23}x_2$$

$$x_4 = a_{34}x_3 + a_{44}x_4$$

$$x_5 = a_{35}x_3 + a_{45}x_4$$



Mason's Gain Formula: The overall system gain is determined by the Mason's gain formula

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

Where, P_k = path gain of Kth forward path

Δ = determinant of the graph = 1 - (sum of loop gains of all individual loops) + (sum of gain products of all possible combinations of two non-touching loops) - (sum of gain products of all possible combinations of three non-touching loops) + ...,

Δ_k = the value of Δ for the part of the graph not touching the K-th forward path

T = overall gain of the system

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 1

Forward Paths:

$$P_1 = X_1 X_2 X_3 X_4 X_5 = a_{12} a_{23} a_{34} a_{45}$$

$$P_2 = X_1 X_2 X_3 X_5 = a_{12} a_{23} a_{35}$$

Loop Gains:

$$L_1 = X_2 X_3 X_2 = a_{23} a_{32}$$

$$L_2 = X_2 X_3 X_4 X_2 = a_{23} a_{34} a_{42}$$

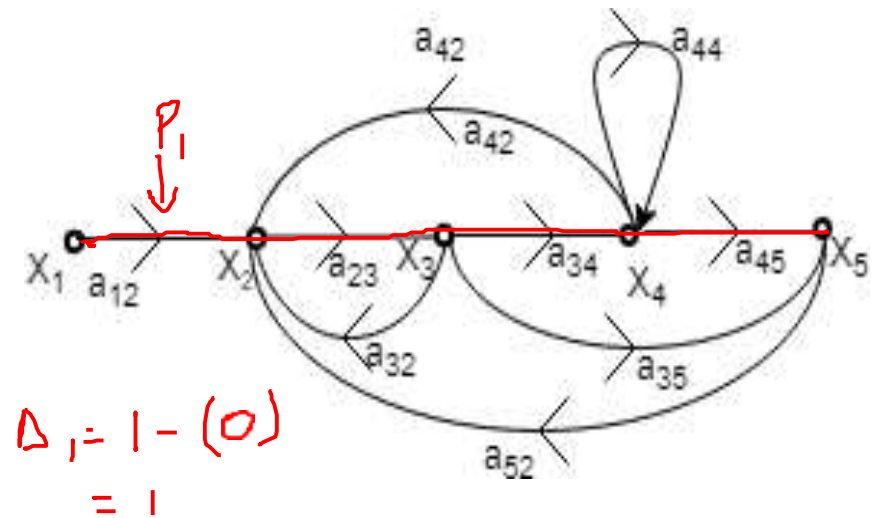
$$L_3 = X_2 X_3 X_4 X_5 X_2 = a_{23} a_{34} a_{45} a_{52}$$

$$L_4 = X_4 X_4 = a_{44}$$

$$L_5 = X_2 X_3 X_5 X_2 = a_{23} a_{35} a_{52}$$

Two Non-Touching loops

$$L_1 L_4 = a_{23} a_{32} a_{44} \quad , \quad L_5 L_4 = a_{23} a_{35} a_{52} a_{44}$$



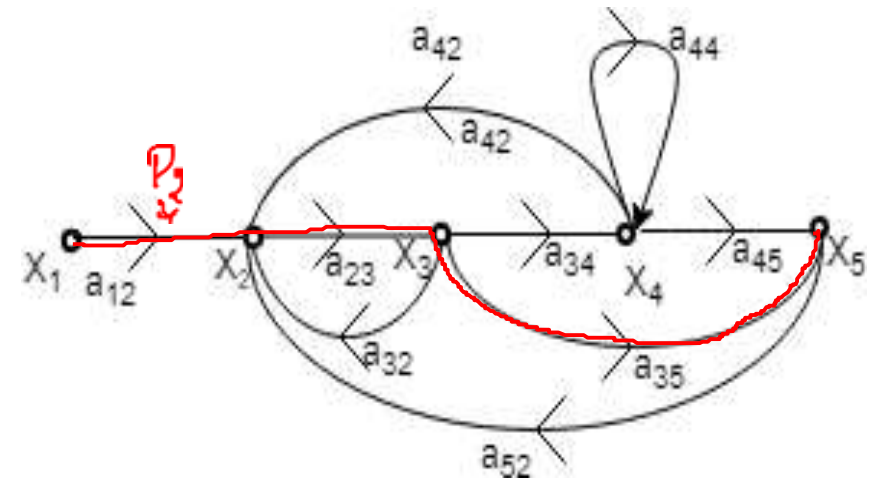
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph - Example

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + (L_1 L_4 + L_5 L_4)$$

$$\Delta_1 = 1 - 0 = 1$$

$$\Delta_2 = 1 - L_4$$



$$T = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta} = \frac{a_{12} a_{23} a_{34} a_{45} + a_{12} a_{23} a_{35} (1 - a_{44})}{1 - (a_{23} a_{32} + a_{23} a_{34} a_{42} + a_{23} a_{34} a_{45} a_{52} + a_{44} + a_{23} a_{35} a_{52}) + (a_{23} a_{32} a_{44} + a_{23} a_{35} a_{52} a_{44})}$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 2

Forward Paths:

$$P_1 = G_1 G_2 G_3 G_4$$

$$P_2 = G_5 G_6 G_7 G_8$$

Loop Gains:

$$L_1 = G_2 H_2$$

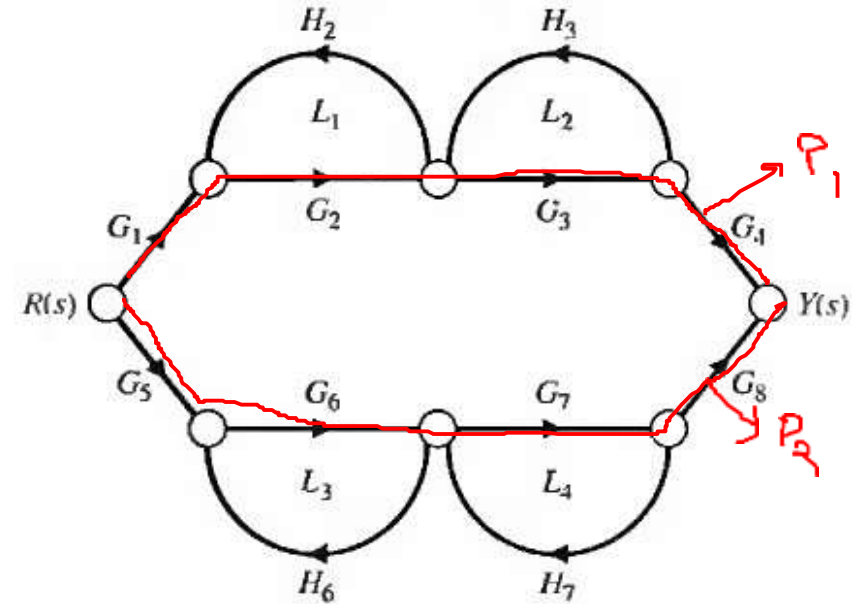
$$L_2 = G_3 H_3$$

$$L_3 = G_6 H_6$$

$$L_4 = G_7 H_7$$

Two Non-Touching loops

$$L_1 L_3, L_1 L_4, L_2 L_3, L_2 L_4$$



$$\Delta_1 = 1 - (L_3 + L_4) + (0)$$

$$\Delta_2 = 1 - (L_1 + L_2) + (0)$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 2



$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_3 + L_1L_4 + L_2L_3 + L_2L_4)$$

$$L_1 = L_2 = 0 \quad \text{and} \quad \Delta_1 = 1 - (L_3 + L_4).$$

$$\Delta_2 = 1 - (L_1 + L_2).$$

$$T = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1G_2G_3G_4(1 - L_3 - L_4) + G_5G_6G_7G_8(1 - L_1 - L_2)}{1 - L_1 - L_2 - L_3 - L_4 + L_1L_3 + L_1L_4 + L_2L_3 + L_2L_4}.$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 3

Forward Paths:

$$P_1 = G_1 G_2 G_3 G_4 G_5 G_6$$

$$P_2 = G_1 G_2 G_7 G_6$$

$$P_3 = G_1 G_2 G_3 G_4 G_8$$

Loop Gains:

$$L_1 = -G_4 H_4 \quad L_5 = -G_2 G_7 H_2$$

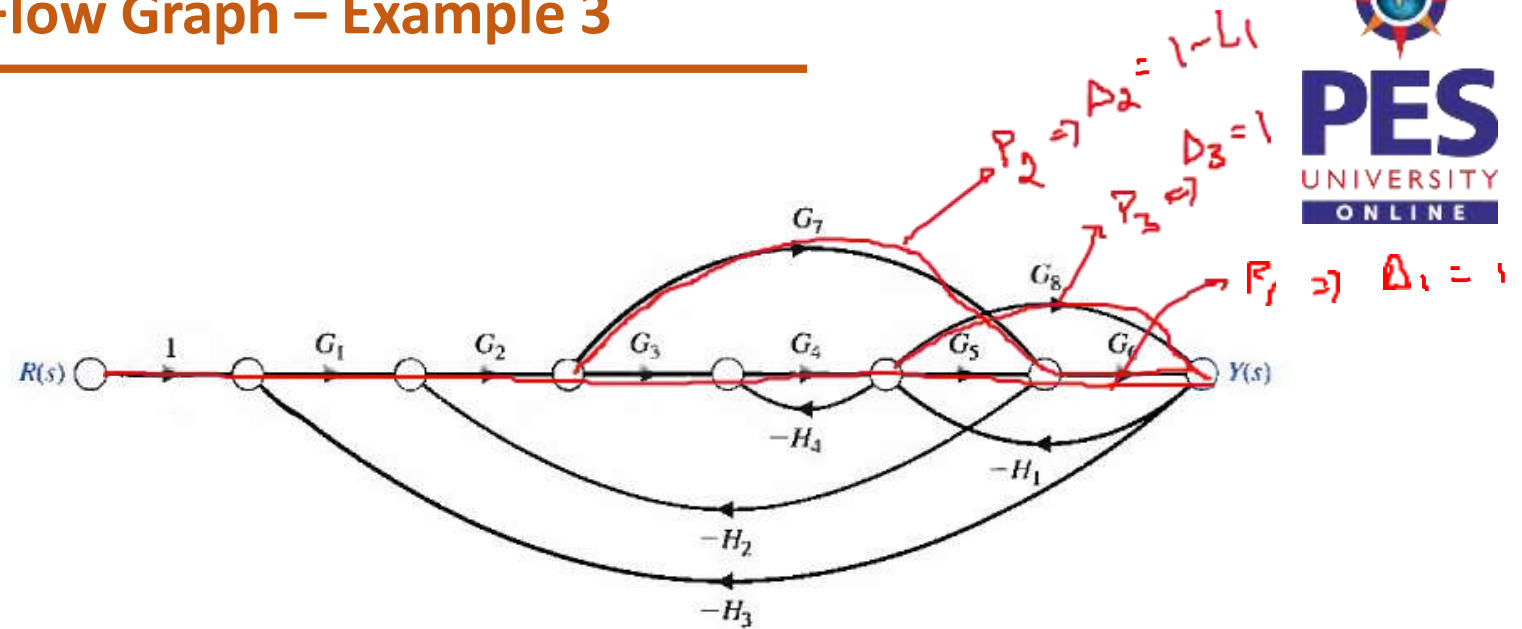
$$L_2 = -G_5 G_6 H_1 \quad L_6 = -G_8 H_3$$

$$L_3 = -G_2 G_3 G_4 G_5 H_2 \quad L_7 = -G_1 G_2 G_7 G_6 H_3$$

$$L_4 = -G_1 G_2 G_3 G_4 G_5 G_6 H_3 \quad L_8 = -G_1 G_2 G_3 G_4 G_8 H_3$$

Two Non-Touching loops

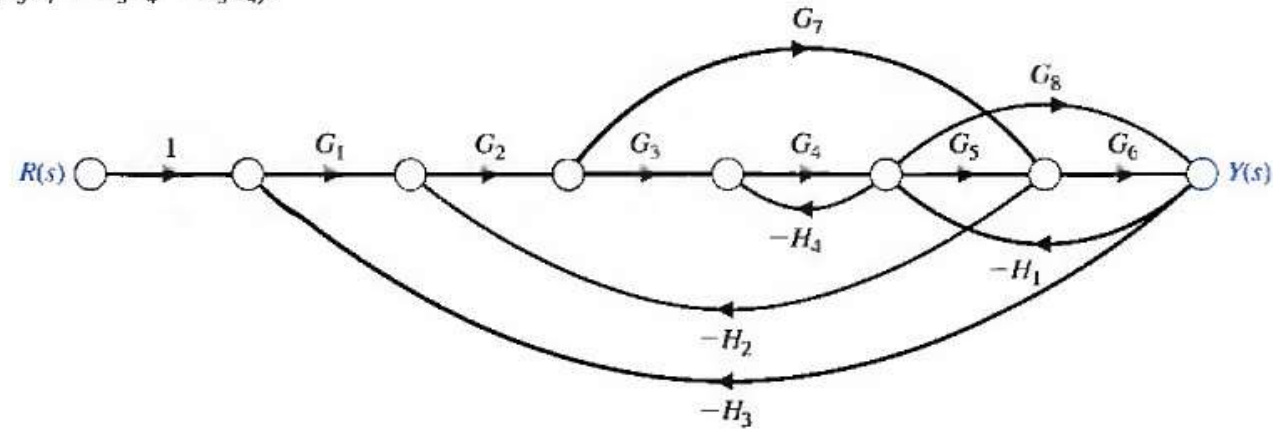
$$L_1 L_7, L_1 L_5, L_5 L_6$$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 2

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6 + L_7 + L_8) + (L_5L_7 + L_5L_4 + L_3L_4).$$



$$\Delta_1 = \Delta_3 = 1 \quad \text{and} \quad \Delta_2 = 1 - L_5 = 1 + G_4H_4.$$

$$T = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

$$T(s) = \frac{Y(s)}{R(s)} = \frac{P_1 + P_2\Delta_2 + P_3}{\Delta}.$$

MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 4

forward paths

$$P_1 = G_1 G_2 G_3 G_4 G_5, \quad \Delta_1 = 1$$

$$P_2 = G_1 G_2 G_7, \quad \Delta_2 = 1 - L_1$$

$$P_3 = G_1 G_6 G_4 G_5, \quad \Delta_3 = 1$$

Loop gains

$$L_1 = -G_4 H_1$$

$$L_2 = -G_2 G_3 G_4 G_5 H_2$$

$$L_3 = -G_2 G_7 H_2$$

$$L_4 = -G_6 G_4 G_5 H_2$$

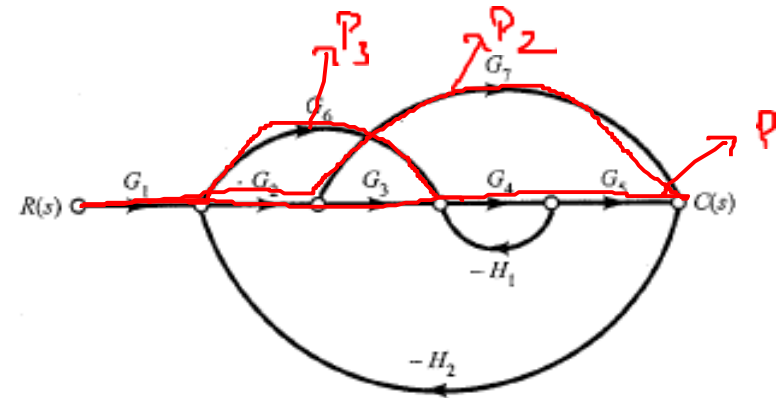
2- non touching loops

$L_1 L_3$

$$\frac{C(s)}{R(s)} = P = \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3)$$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + L_1 L_2$$

$$= \frac{G_1 G_2 G_3 G_4 G_5 + G_1 G_6 G_4 G_5 + G_1 G_2 G_7 (1 + G_4 H_1)}{1 + G_4 H_1 + G_2 G_7 H_2 + G_6 G_4 G_5 H_2 + G_2 G_3 G_4 G_5 H_2 + G_4 H_1 G_2 G_7 H_2}$$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 5

$$P_1 = G_1 G_2 G_3$$

$$L_1 = G_1 G_2 H_1$$

$$L_2 = -G_2 G_3 H_2$$

$$L_3 = 1$$

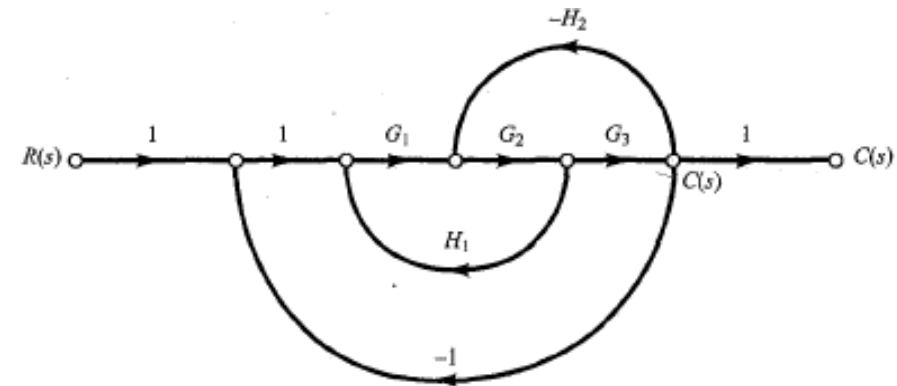
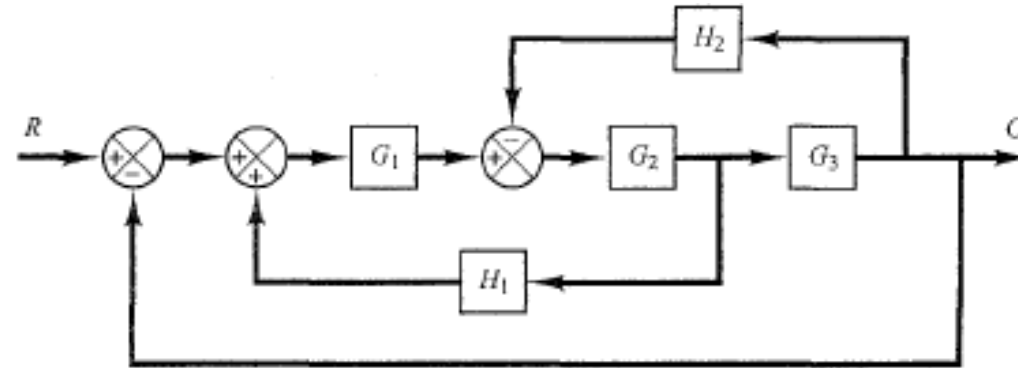
$$\Delta = 1 - (L_1 + L_2 + L_3)$$

$$= 1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3$$

$$\Delta_1 = 1$$

$$\frac{C(s)}{R(s)} = P = \frac{P_1 \Delta_1}{\Delta}$$

$$= \frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$$

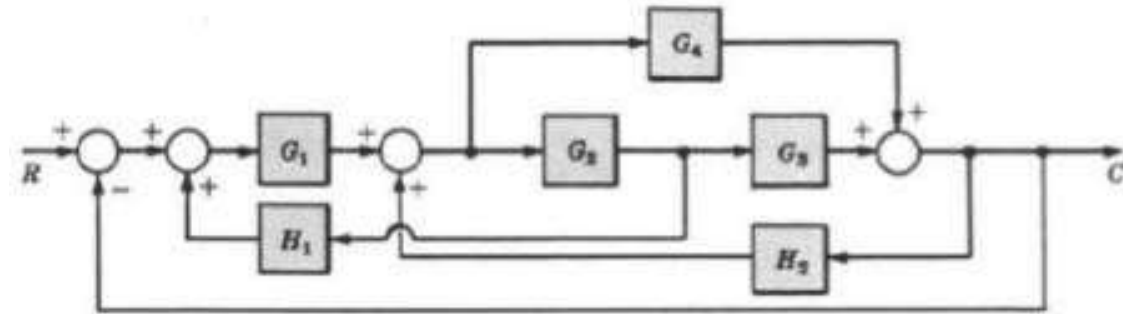


MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Signal Flow Graph – Example 6

The two forward path gains are

$$P_1 = G_1 G_2 G_3 \text{ and } P_2 = G_1 G_4$$



$$L_1 = G_1 G_2 H_1$$

$$L_2 = G_2 G_3 H_2$$

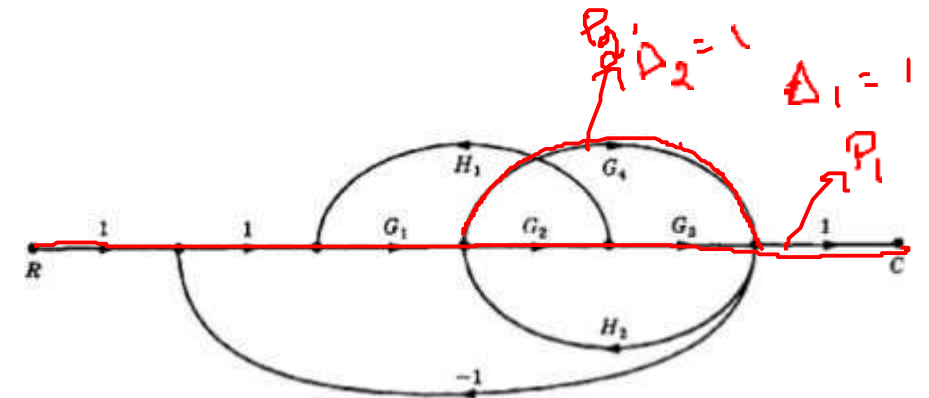
$$L_3 = -G_1 G_2 G_3$$

$$L_4 = G_4 H_2$$

$$L_5 = -G_1 G_4$$

$$T = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

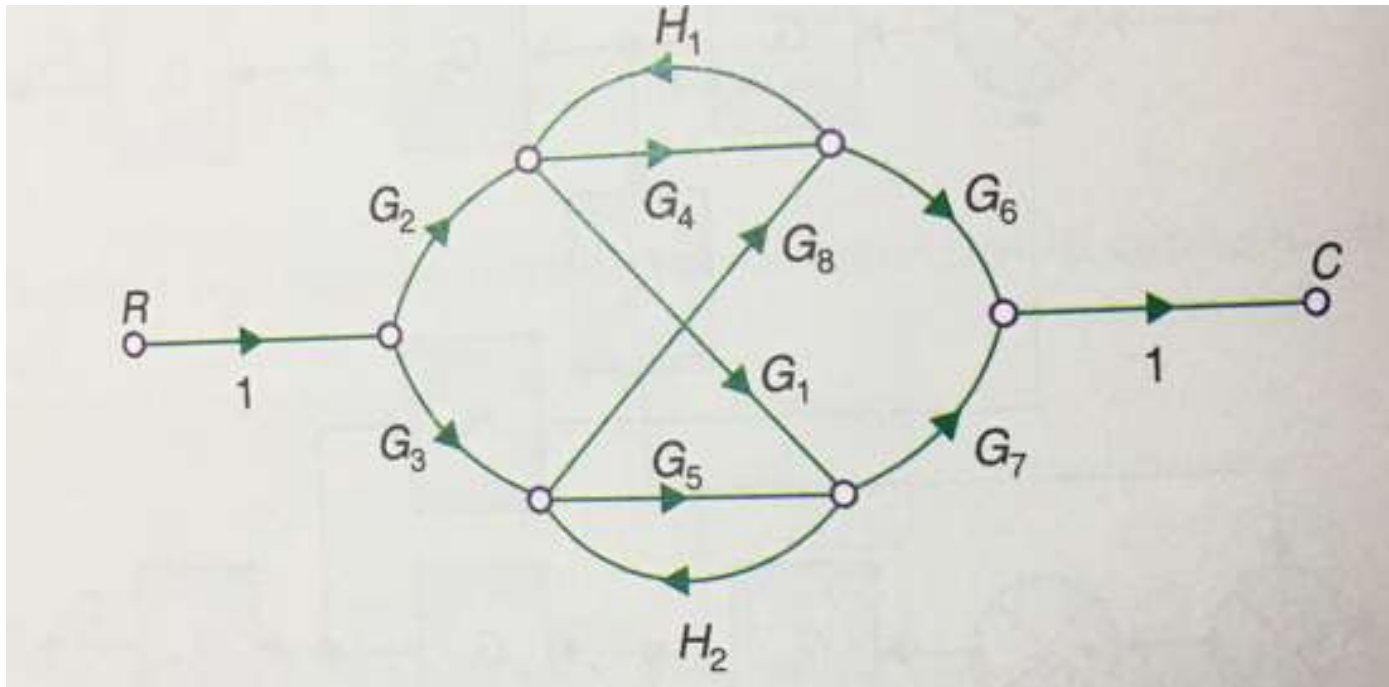
$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5)$$



MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram – Example 4

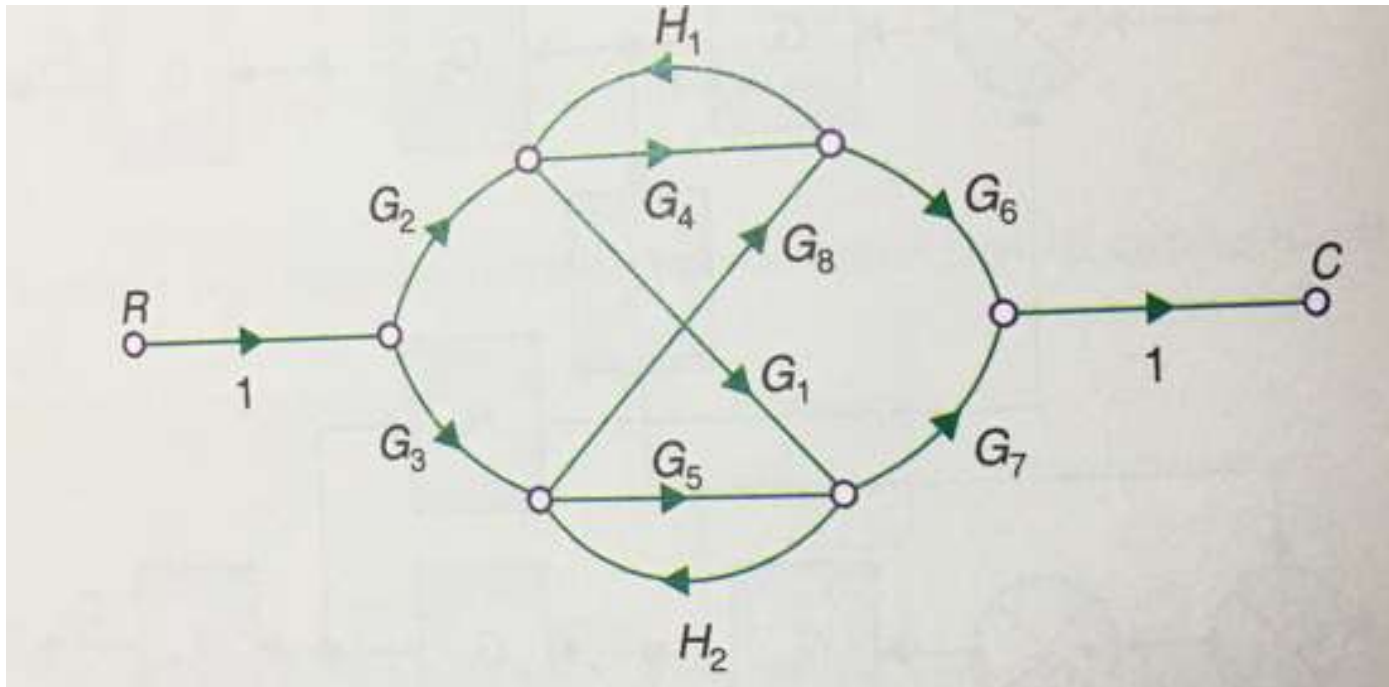
Find overall transfer function of the system as shown in figure using Mason's Gain Formula

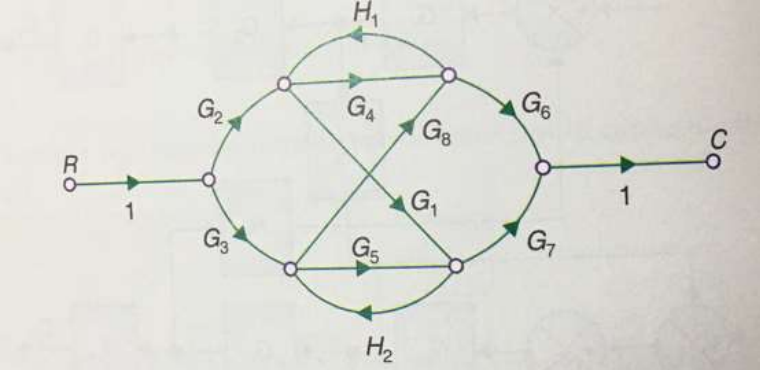


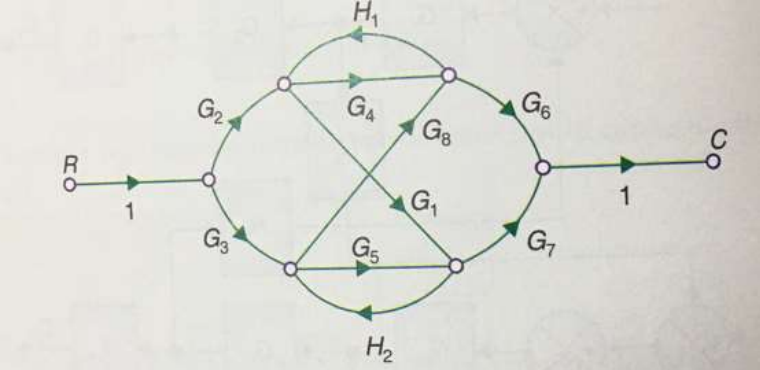
MATHEMATICAL MODELS OF PHYSICAL SYSTEMS

Transfer Functions using Block Diagram – Example 4

Find overall transfer function of the system as shown in figure using Mason's Gain Formula









THANK YOU

Dr. Anuradha M

Department of Electronics and Communication
Engineering

anuradha@pes.edu

+91 80 2672 1983 Extn: 774